industrial automation 2026

roberto sacile

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Assignment 1 – Modeling and Control of a Cascaded Three-Tank Aeration System

Process description

Consider the industrial process shown schematically as a cascade of three interconnected tanks. The liquid flows from Tank 1 to Tank 2 and from Tank 2 to Tank 3. Tank 2 is equipped with an aeration system, while Tank 3 is equipped with a discharge pump and a recycle pump sending part of the outlet flow back to Tank 1.

The system is intended to represent a simplified water treatment and oxygenation process. The objective is to regulate liquid levels and oxygen concentration while following prescribed time-varying references.

State variables, outputs, and control inputs

Assume the following state variables:

$$x(t) = \begin{bmatrix} h_1(t) \\ h_2(t) \\ h_3(t) \\ c_2(t) \\ c_3(t) \end{bmatrix}$$

where:

- $h_1(t)$, $h_2(t)$, $h_3(t)$ are the liquid levels in Tanks 1, 2, and 3 [m];
- $c_2(t)$ is the dissolved oxygen concentration in Tank 2 [mg/L];
- $c_3(t)$ is the dissolved oxygen concentration in Tank 3 [mg/L].

The measured outputs are:

$$y(t) = \begin{bmatrix} y_1(t) \\ y_2(t) \\ y_3(t) \end{bmatrix} = \begin{bmatrix} h_2(t) \\ h_3(t) \\ c_3(t) \end{bmatrix}$$

corresponding respectively to:

- a level sensor in Tank 2;
- a level sensor in Tank 3;
- a dissolved oxygen sensor in Tank 3.

The manipulated inputs are:

$$u(t) = \begin{bmatrix} u_a(t) \\ u_r(t) \\ u_d(t) \end{bmatrix}$$

where:

- $u_a(t)$ is the aeration input in Tank 2;
- $u_r(t)$ is the recycle flow command from Tank 3 to Tank 1;
- $u_d(t)$ is the discharge flow command at the outlet of Tank 3.

At least two manipulated inputs must be used in the control design.

Given parameters

Assume the following numerical parameters:

- tank cross-sectional areas:

$$A_1 = 1.2 \text{ m}^2, \qquad A_2 = 1.0 \text{ m}^2, \qquad A_3 = 1.4 \text{ m}^2;$$

- inlet flow to Tank 1:

$$q_{\text{in}} = 0.18 \text{ m}^3/\text{s};$$

- oxygen concentration at the inlet:

$$c_{\text{in}} = 1.0 \text{ mg/L};$$

- hydraulic coefficients between tanks:

$$k_{12} = 0.22, \qquad k_{23} = 0.20;$$

- maximum aeration transfer coefficient:

$$k_a = 0.9;$$

- saturation oxygen concentration in Tank 2:

$$c_{\text{sat}} = 8.0 \text{ mg/L};$$

- oxygen decay coefficients:

$$\lambda_2 = 0.08 \text{ s}^{-1}, \qquad \lambda_3 = 0.05 \text{ s}^{-1}.$$

Assume the manipulated variables satisfy the constraints

$$0 \le u_a(t) \le 1, \qquad 0 \le u_r(t) \le 0.08 \text{ m}^3/\text{s}, \qquad 0.05 \le u_d(t) \le 0.25 \text{ m}^3/\text{s}.$$

Qualitative modeling guidelines

The student must derive the model from physical considerations, using mass balances and concentration balances.

The following qualitative interactions must be included:

1. The level in each tank varies according to the difference between inflows and outflows.
2. The flow from Tank 1 to Tank 2 depends on the level difference $h_1 - h_2$.
3. The flow from Tank 2 to Tank 3 depends on the level difference $h_2 - h_3$.
4. The recycle flow decreases the level in Tank 3 and increases the level in Tank 1.
5. The oxygen concentration in Tank 2 increases because of aeration and is affected by inflow, outflow, and dilution.
6. The oxygen concentration in Tank 3 depends on the incoming oxygen from Tank 2, on the discharge flow, and on the recycle flow.
7. The state equations must be coupled, i.e. each state should affect the dynamics of other states.

You may assume nonlinear hydraulic laws of the form

$$q_{12}(t) = k_{12}\sqrt{\max(h_1(t) - h_2(t),\, 0)}$$

and

$$q_{23}(t) = k_{23}\sqrt{\max(h_2(t) - h_3(t),\, 0)}.$$

Reference trajectories

The plant outputs must follow the following desired trajectories over the time interval $t \in [0, 150]$ s:

$$h_{2,\text{ref}}(t) = 0.80 + 0.08\sin\left(\frac{2\pi}{60}t\right) \quad [\text{m}]$$

$$h_{3,\mathrm{ref}}(t) = 0.90 + 0.06 \sin\left(\frac{2\pi}{90}t + \frac{\pi}{4}\right) \quad [\mathrm{m}]$$

$$c_{3,\mathrm{ref}}(t) = 5.50 + 0.70\left(1 - e^{-t/40}\right) + 0.20 \sin\left(\frac{2\pi}{75}t\right) \quad [\mathrm{mg/L}]$$

These references are chosen so that the operating conditions remain physically meaningful.

Tasks

1. Nonlinear continuous-time model

Derive a nonlinear continuous-time state-space model of the plant in the form

$$\dot{x}(t) = f(x(t), u(t)), \qquad y(t) = g(x(t)).$$

The model must include:

- three level balance equations;
- two oxygen balance equations;
- at least two manipulated inputs;
- nonlinear coupling among the state variables.

Clearly define all intermediate flows and explain the assumptions used.

2. Continuous-time PID control design

Using the nonlinear continuous-time model, propose a control strategy based on PID controllers. Use at least two PID controllers.

A possible choice is:

- one PID controller acting on $u_a(t)$ for tracking $c_{3,\mathrm{ref}}(t)$;
- one PID controller acting on $u_d(t)$ for tracking $h_{3,\mathrm{ref}}(t)$;
- optionally, one additional PID controller acting on $u_r(t)$ to improve the regulation of $h_2(t)$ around $h_{2,\mathrm{ref}}(t)$.

Students are free to propose a different pairing if properly justified.
Write the PID laws in continuous time and explain the role of each term.

3. Digital PID with ZOH applied to the continuous-time plant

Assume now that the control algorithm is implemented digitally with sampling time T_s. The plant remains continuous-time and the control input is applied through a zero-order hold (ZOH).

Choose an initial sampling time

$$T_s = 0.5 \text{ s}$$

and write the discrete-time PID control law explicitly. For each control loop, define:

- the tracking error;
- the proportional term;
- the integral update;
- the derivative approximation.

Then connect the discrete PID controller to the continuous-time nonlinear plant through a ZOH block.

4. Euler discretization of the nonlinear model

Discretize the continuous-time nonlinear model using the forward Euler method with sampling time T_s. Obtain a nonlinear discrete-time model of the form

$$x_{k+1} = F(x_k, u_k), \qquad y_k = G(x_k).$$

Using this discrete-time nonlinear model, redesign or retune the PID controllers in discrete time and simulate the system.

Compare the results with those obtained in Task 3.

5. Equilibrium point and linearization

Determine an equilibrium point $(x^\star, u^\star)$ compatible with the assigned operating regime. Since the references are time-varying, choose an equilibrium corresponding to representative average values of the references.

Linearize the nonlinear continuous-time model around $(x^\star, u^\star)$ and derive a linear model of the form

$$\delta\dot{x}(t) = A\,\delta x(t) + B\,\delta u(t), \qquad \delta y(t) = C\,\delta x(t) + D\,\delta u(t).$$

Show the matrices A, B, C, and D.

6. Discretization of the linearized model

Discretize the linearized model using the same sampling time T_s and obtain

$$\delta x_{k+1} = A_d\,\delta x_k + B_d\,\delta u_k, \qquad \delta y_k = C_d\,\delta x_k + D_d\,\delta u_k.$$

Then propose again a discrete-time PID control design based on this linearized-discrete model.

Compare the behavior of:

- discrete PID + ZOH + continuous nonlinear plant;
- discrete PID + Euler-discretized nonlinear plant;
- discrete PID + discretized linearized plant.

7. Effect of the sampling time

Repeat the digital control analysis for the following sampling times:

$$T_s \in \{0.2\text{ s},\ 0.5\text{ s},\ 1.0\text{ s},\ 2.0\text{ s}\}.$$

Discuss how the sampling time affects:

- tracking accuracy;
- transient response;
- oscillations;
- control effort;
- numerical approximation quality;
- possible closed-loop instability or performance degradation.

Initial conditions

Use the following initial conditions:

$$h_1(0) = 0.75\text{ m}, \qquad h_2(0) = 0.70\text{ m}, \qquad h_3(0) = 0.85\text{ m},$$

$$c_2(0) = 4.00\text{ mg/L}, \qquad c_3(0) = 4.20\text{ mg/L}.$$

What must be submitted

The report must contain:

1. derivation of the nonlinear continuous-time model;
2. definition of states, inputs, outputs, and sensors;
3. continuous-time PID control laws;
4. discrete-time PID laws with ZOH applied to the continuous-time plant;
5. Euler discretization of the nonlinear model;
6. discrete-time PID design for the Euler-discrete model;
7. equilibrium computation and linearization;
8. discretization of the linearized model;
9. discrete-time PID design for the linearized-discrete model;
10. simulation results and comparison for different sampling times.

Minimum simulation plots

Include at least the following figures:

- $h_2(t)$ and $h_{2,\mathrm{ref}}(t)$;
- $h_3(t)$ and $h_{3,\mathrm{ref}}(t)$;
- $c_3(t)$ and $c_{3,\mathrm{ref}}(t)$;
- control signals $u_a(t)$, $u_r(t)$, and $u_d(t)$;
- comparison among the three modeling/control approaches;
- comparison for different values of T_s.

Optional bonus points

Bonus points may be assigned if the student includes one or more of the following:

- actuator saturation handling and (if you have a background on this) anti-windup strategies;
- measurement noise;
- disturbance on q_{in};
- discussion of multivariable interaction and loop coupling;
- sensitivity of the linearized model to the chosen equilibrium point.

Hints

- This text has been generated by the professor with the help of AI. If something is unclear please ask the professor
- A physically reasonable and internally consistent model is preferred over an unnecessarily complicated one.
- Clearly justify every modeling assumption.
- When choosing the equilibrium point, use representative average values of the assigned reference trajectories.
- When comparing different sampling times, comment not only on tracking performance but also on numerical and implementation issues.